Discrete Mathematics

Introduction to Discrete Mathematics

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UNIT 1 – Discrete Theory, Relations, and Functions

1. INTRODUCTION TO DISCRETE MATHEMATICS

1.1 OVERVIEW AND SCOPE

DISCRETE MATHEMATICS IS THE BRANCH OF MATHEMATICS THAT DEALS WITH FINITE OR COUNTABLY INFINITE STRUCTURES. UNLIKE CONTINUOUS MATHEMATICS (LIKE CALCULUS), WHERE VALUES CAN VARY SMOOTHLY, DISCRETE MATHEMATICS STUDIES OBJECTS THAT HAVE DISTINCT AND SEPARATE VALUES.

Core Idea:
If you can count the elements one by one, it's likely part of discrete mathematics.
Key Areas Covered in Discrete Mathematics:

- Set Theory – Collection of distinct elements and their operations.
- Logic – Study of reasoning, propositions, and truth values.
- Relations and Functions – Ways of associating elements of sets.
- Combinatorics – Counting, arrangements, and probability.
- Graph Theory – Networks of connected nodes and edges.
- Number Theory – Properties and relationships of integers

Why It Matters:
Discrete mathematics forms the mathematical foundation for most of modern computer science, digital electronics, and algorithm design.
Scope of Discrete Mathematics:

- Mathematical Logic: Basis for reasoning in programming and algorithms.
- Data Structures: Arrays, stacks, queues, linked lists rely on discrete principles.
- Cryptography: Securing information through discrete number theory.
- Network Design: Graph theory models the internet and social networks.
- Algorithm Analysis: Counting steps, efficiency, and complexity.

1.2 APPLICATIONS IN COMPUTER SCIENCE

DISCRETE MATHEMATICS IS LIKE THE LANGUAGE OF COMPUTER SCIENCE. MANY COMPUTING CONCEPTS DIRECTLY COME FROM IT.

1. Computer Algorithms
   - Sorting algorithms (Quick Sort, Merge Sort) rely on combinatorics and logic.
   - Searching algorithms use discrete structures like trees and graphs.
2. Data Structures
   - Arrays, stacks, queues, linked lists → built from set theory and logic.
3. Cryptography and Security
   - Public-key encryption (RSA) is based on number theory and prime factorization.
4. Database Design
   - SQL queries use set theory for UNION, INTERSECT, and JOIN.
5. Artificial Intelligence
   - Knowledge representation uses predicate logic.
   - Decision-making in AI uses inference rules from discrete logic.
6. Computer Networks
   - Graph theory models network topologies and routing algorithms.
7. Digital Logic Design
   - Logic gates and circuits are based on Boolean algebra.

Real-World Example:
When you use Google Maps, the app finds the shortest path between two points. This uses Dijkstra's algorithm, which is based on graph theory — a core area of discrete mathematics.

1.3 CONTINUOUS VS. DISCRETE STRUCTURES

Continuous Structures:
- Deal with values that change smoothly.
- Examples: Real numbers, curves, calculus.
- Used in physics, engineering, signal processing.
- Analogy: Like water flowing — it can take any value in between.

Discrete Structures:
- Deal with separate, distinct values.
- Examples: Integers, graphs, finite sets, logic.
- Used in computer science, cryptography, networking.
- Analogy: Like LEGO blocks — each piece is separate and countable.

2.Set Theory

2.1 SETS & TYPES OF SET

THE THEORY OF SETS WAS DEVELOPED BY GERMAN MATHEMATICIAN GEORGE CANTER (1845-1918 A.D).

Definition:
A set is a well-defined collection of distinct objects, called elements or members.
(i) The objects in a set are called its members or elements.
(ii) We denote sets by capital letters A, B, C, X, Y, Z, etc.
(iii) The elements of a set are represented by small letters a, b, c, x, y, z, etc.
If "a" is an element of a set A, we write a A i.e a belongs to A or is an element of A. If a does not belong to A, it can be written as a A.

Illustration:

- The collections of all natural numbers denoted by N.
- The collections of all vowels in the English alphabet is the set containing given elements, namely a, e, i, o, u.
- The solution of the equation x - 5x + 6 = 0 i.e, 2 and 3.
- The rivers of India.
- The collection of all even numbers less then 8, is the set containing the numbers 2,4,6.
- The collection of all integers, all rational numbers and all real numbers are denoted by Z, Q and R.

Representation of Sets:
There are two methods of representing set
a) Roaster or tabular form:
In this form, all the elements of a set are listed within braces and are separated by commas.
Example :Write the following sets in the roster form.
(i) A = Sets of all factors of 12
(ii) B = set of all positive integers less than 20
(iii) C = Set of all letters in the word 'MATHEMATICS'
Solution :
(i) All factors of 12 are 1,2,3,4,56,12
A = {1,2,3,4,6,12}
(ii) All even positive integers less than 20 are 2,4,6,8,10,12,14,16,18
B = {2,4, 6,8,10,12,14,16,18}
(iii) The set of all letters in the word MATHEMATICS is given by
C = {M, A, T, H, E, I, C, S}
(b) Rule Method or Set Builder Form
In this form, we list the property satisfied by all the elements in the set. It can be written as
{x : s satisfied the property}
i.e, The set of all those x such that each x has properties P.
Example 2: Write the set A = {1, 2, 3, 4, 5, 6, 7, 8} in the set-builder form.
Solution :We have A = Set of all natural numbers less than 9. Thus
A = {x : x N and x<9}

Types of Sets

(i) Empty Set :

A set containing no element is called the empty set or the null set or the void set, denoted by O or { }. A set which has at least one element is called a non-empty set.

Illustrations :

(i) $\{ x : x \quad N \text{ and } 2<x<3\} = O$

(ii) $\{ x : x \quad = 9 \text{ and } x \text{ is an even integer}\} = O$

(ii) Singleton Set :

A set containing exactly one element is called singleton set.

Illustrations :

(i) {0} is a singleton set whose element is 0.

(ii) $\{ x : x \quad N^2 \text{ and } x \quad - 9 = 0\} = \{3\}$ which is a singleton set.

(iii) Finite and Infinite Set :

Set is said to be finite if it consists of only finite number of elements. A set which is not finite is called an infinite set. The number of distinct elements in a finite set A is denoted by n(A).

Illustrations of Finite Sets

(i) Let A = {2, 4, 6, 8, 10}. Then, A is finite set and n(A) = 5.

(ii) The set of all persons on earth is a finite set.

(iii) The set of all animals on earth is finite set.

Illustrations of Infinite Sets

(i) The set of all points on the arc of a circle is an infinite set.

(ii) The set of all positive integer multiples of 5 us an infinite set.

(iii) N = set of all natural numbers = {1, 2, 3, 4, 5, 6............}

(iv) P = set of all prime numbers = {2, 3, 5, 11, 13..................}

(iv) Equal Sets :

Two sets A and B are said to be equal, if they have exactly the same elements and we write, A = B. If the sets are not equal then $A \neq B$.

(i) The elements of a set may be listed in any order.

Thus, {1, 3} = {3, 1} = {3, 1} etc.

(ii) The repetition of elements in a set has no meaning.

Thus {1, 3} = {1, 1, 3, 2} = {1, 1, 2, 3, 3} etc.

(v) Equivalent Sets :

Two finite sets A and B are said to be equivalent if n(A) = n(B).

Illustration: Let A = {2, 3, 4}, B = {7, 8, 9}

Then n(A) = n(B) = 3

So, A and B are equivalent. But $A \neq B$.

Cardinal Number of Set

The number of distinct elements contained in a finite set A is called the cardinal number of A, and is denoted by n(A).

Subsets

A set A is said to be subset of set B, if every element of A is also an element of B and we write, $A \subseteq B$.

Super Set

If $A \subseteq B$ then B is called a super set of A and we write $B \supseteq A$.

Proper Subset

If $A \subseteq B$ and , then A is called a proper subset of B and we write $A \subset B$.

Power Set

The set of all subsets of a given set A is called power set of A and denoted by P(A).

If n(A) = m then, $n[P(A)] = 2^m$

2.2 Relations & Types of Relations

Relation in Mathematics is defined as the relationship between two sets. If we are given two sets set A and set B and set A has a relation with set B then each value of set A is related to a value of set B through some unique relation. Here, set A is called the domain of the relation, and set B is called the range of the relation.

For example, if we are given two sets,

- Set A = {1, 2, 3, 4} and Set B = {1, 4, 9, 16}

The relation R from set A to set B is defined by the rule: $y = x2: y \in B, x \in A\}$

Relation is defined as the relation between two different sets of information. Suppose we are given two sets containing two different values then a relation defined such that it connects the value of the first set with the value of the second set is called the relation.
Suppose we are given a set A that contains the name of girls of a class and another set B that includes the height of girls then a relation connects set A with set B. In mathematical terms, we can say that,

"A set of ordered pairs is a relation"

Example of a Relation

Examples of relation in mathematics include,

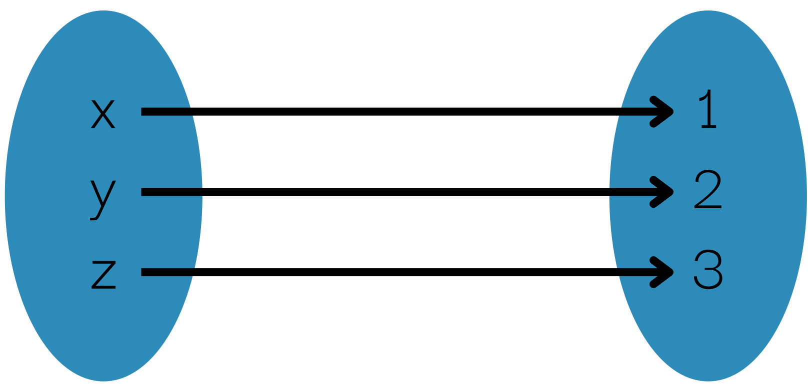
Suppose there are two sets X = {4, 36, 49, 50} and Y = {1, -2, -6, -7, 7, 6, 2}. A relation R states that

”(x, y) is in the relation R if x is a square of y” can be represented using ordered pairs,

- R = {(4, -2), (4, 2), (36, -6), (36, 6), (49, -7), (49, 7)}

Also, the image added below shows two sets A and B, and the relation between them,

- Set A = {x, y, z}
- Set B = {1, 2, 3}



Types of Relation

Various types of relations defined in mathematics are,

- Reflexive Relation
- Symmetric Relation
- Transitive Relation

Now let's learn about them in detail.

(i) Reflexive Relation

A relation R on a set A is called reflexive if (a, a) $\in$ R holds for every element a$\in$ A . i.e. if set A = {a, b} then R = {(a, a), (b, b)} is reflexive relation.

For example, A = {2, 3} then the reflexive relation R on A is,

- R = {(2, 2), (3, 3)}

(ii) Symmetric Relation

A relation R on a set A is called symmetric if (b, a) $\in$ R holds when (a, b) $\in$ R i.e. The relation R = {(a, b), (b, a)} is a reflexive relation on (a, b)

For example, A = {2, 3} then symmetric relation R on A is,

- R = {(2, 3), (3, 2)}

(iii) Transitive Relation

A relation R on a set A is called transitive if (a, b) $\in$ R and (b, c) $\in$ R then (a, c) $\in$ R for all a,b,c $\in$ A i.e.

For example, set A = {1, 2, 3} then transitive relation R on A is,

- R = {(1, 2), (2, 3), (1, 3)}

UNIT 2 - Propositional Logic

MATHEMATICAL LOGICS

1.1 Basics of Logics

Mathematical logic deals with the logic in mathematics. Mathematical logic operators and laws define various statements in their mathematical form. In this article, we will explore mathematical logic along with the mathematical logic operators and types of mathematical logic. We will also solve some examples related to mathematical logic.

What is Mathematical Logic?

The study of mathematical logic in mathematics is called mathematical logic. The basic mathematical logic used are the conjunction ($\wedge$), disjunction ($\vee$), and negation (¬) and implication.

Mathematical Logic Operators

The basic mathematical logic operators are:

- Conjunction : In mathematical logic conjunction of two statements results in true when both the statements are true otherwise false. Conjunction is also known as AND operator and is represented by $\wedge$.
- Disjunction : In mathematical logic disjunction of two statements results in false if both the statements are false otherwise true. Disjunction is also known as OR operator and is represented by $\vee$.
- Negation : In mathematical logic negation of two statements results in the not of the given statement i.e., if the statement is true it results in false and if the statement is false it results in true. Negation is also known as NOT operator and is represented by ~ or ¬.
- Implication : In mathematical logic implication of two statements results in false if the first statement is true and second statement is false otherwise true. Implication is also known as conditional operator and is represented by → or $\Rightarrow$. Implication X→Y is read as If X and then Y.

Mathematical Logic Truth Table

The truth table in mathematical logic is a table which takes inputs and provides output when a logic is applied to it. The truth table for different mathematical logic operators are given below.

Negation

The truth table for negation is given below.

T	**~x**
T	F
F	T

Conjunction

The truth table for conjunction is given below.

A	**B**	**A⋀B**
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

The truth table for disjunction is given below.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Implication

The truth table for implication is given below.

A	B	A B
T	T	T
T	F	F
F	T	T
F	F	T

Mathematical Logic Solved Examples

Example 1: Consider the statement $x < 5 \rightarrow x - 2 < 5$ is true or false?

Solution:

If $x < 5$ is true then, $x - 2 < 5$ is also true.

$T \rightarrow T$ is true

So, the given statement $x < 5 \rightarrow x - 2 < 5$ is true.

1.2 Applications of Logics

Applications of Propositional Logic

A proposition is an assertion, statement, or declarative sentence that can either be true or false but not both. For example, the sentence "Ram went to school." can either be true or false, but the case of both happening is not possible. So we can say, the sentence "Ram went to school." is a proposition. But, the sentence "N is greater than 100" is not a proposition as we cannot state whether it is true or false unless the value of N is given. A few more examples of propositions are "12 - 10 = 3", "The library is open.", etc.

The area of mathematical logic that deals with propositions is called Propositional Logic or Propositional Calculus. It is also known as sentential logic or sentential calculus. It studies the logical values of propositions when taken as a whole and logical relationships when connected using logical connectives(such as logical AND, logical OR, etc.).

1.3 Importance of Propositional Logic

Propositional Logic plays an important role in computer science as well as in a person's daily life. The main benefits of studying and using propositional logic are that it prevents us from making inconsistent inferences and incautious decisions. It incorporates reasoning and thinking abilities in one's daily life.

Applications of Propositional Logic

In the computer science field, propositional logic has a wide variety of applications and hence is very important. It is used in system specifications, circuit designing, logical puzzles, etc. Apart from this, it can also be used in translating English sentences to mathematical statements and vice-versa. Let us look at this vast variety of applications in detail.

1) Translating English Sentences into logical statements
Like any other human language, English sentences can be ambiguous. This ambiguity might lead to uninformed decision-making and other fatal errors. To remove this ambiguity, we can translate these English sentences into logical expressions with the help of Propositional Logic. Note that sometimes this may include making a few assumptions based on the sentence's intended meaning.

Example: Given a sentence "You can purchase this book if you have $20 or $10 and a discount coupon." Now, this is a bit complex to be understood at once. So we translate this into a logical expression that will make it simple to understand. Let a, b, c, and d represent the sentences "You can purchase this book.", "You have $20.", "You have $10.", and "You have a discount coupon." respectively. Then the given sentence can be translated to (b $\vee$ (c $\wedge$ d) -> a, which simply means that "if you either have $20 or $10, along with a discount coupon, then you can purchase the book."

2) System Specifications
When developing/manufacturing a system(Software or Hardware), the developers/manufacturers have to meet certain needs and specifications, which are usually stated in English. But as English sentences can be ambiguous, developers/engineers translate these system specifications into logical expressions to state specifications rigorously and unambiguously.

Example: Let a, b, c, and d represent the sentences "The computer is within the local network.", "The computer has a valid login id.", "The computer is under the use of administrator.", and "Internet is accessible to the computer." So the complex sentence, "If the computer is within the local network or it is not within the local network but has a valid login id or it is under the use of administrator, then the Internet is accessible to the computer." can be expressed as :
$(a \lor (\neg a \land b) \lor c) \to d$.

3) Logical Puzzles

Puzzles that are solved using reasoning and logic are called logical puzzles. They can be used for brain exercises, recreational purposes, and for testing a person's reasoning capabilities. Solving such puzzles is generally tricky, but it can be done easily using propositional logic. Some of the famous logic puzzles are the muddy children puzzle, Smullyan's puzzles about knights and knaves, etc.

Example:

Problem Statement: There is an island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B. Determine what are A and B if A says "B is a knave" and B says "Both of us are of same types"?

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively, so $\neg p$ and $\neg q$ are the statements that A is a knave and B is a knave, respectively. Let's assume A is a knight, i.e., p is true. So A is telling the truth, which means $\neg q$ is true. Now as B is a knave whatever it is saying is a lie, i.e., $(p \land q) \lor (\neg p \land \neg q)$ is false, which simply means if either of them is a knave then the other one is a knight or vice-versa. Now, as per the assumption, this statement is true. So we can conclude that A is a knight and B is a knave.

4) Boolean Searches

Another important application of propositional logic is Boolean searches. These searches use techniques from propositional logic. Logical connectives are used extensively in searches of large collections of information, such as indexes of Web Pages. In Boolean searches, the connective AND is used to find records that contain both of the two terms, the connective OR is used to find records that have either one or both of the terms, and the connective NOT, also written as AND NOT, is used when we need to exclude a particular search term.

Example: Some web search engines support web page searching that uses Boolean techniques. For instance, if we want to look for web pages about hiking in West Alps, then we can look for pages matching WEST AND ALPS AND HIKING. And if we want web pages about hiking in the Alps, but not in West Alps, then we can search for pages matching the record HIKING AND ALPS AND NOT(WEST AND ALPS).

5) Logic/Computer Circuits

Logic gates or circuits are electronic devices that implement Boolean functions, i.e. it does a logic operation on one or more bits of input and gives a bit as an output. They are the basic building blocks of any digital system. The relationship between the input and output is based on a certain propositional logic.

Example: Logic gates are named as AND gate, OR gate, NOT gate, etc. based on the names of logical connectives AND, OR, NOT, etc. The output truth values for the given input bits for these gates are the same as that those returned by the logical connectives.

6) Inference and Decision Making

Propositional Logic is widely used in the making rules of inference and decision making. These rules of inferences can then be used to build arguments. When several premises are given, it is hard to tell if a given argument is valid. Thus, we use these rules of inference to validate an argument and make a decision.

Example: We can prove that the following premises build up a valid argument using the rule of inference.

"If today is Tuesday, I have a test in English or Science. If my English Professor is absent, then I will not have a test in English. Today is Tuesday and my English Professor is absent. Therefore I have a test in Science."

T: Today is Tuesday

E: I have a test in English

S: I have a test in Science

A: My English Professor is absent

7) Artificial Intelligence - Fuzzy Logic

AI algorithms make use of fuzzy logic. In fuzzy logic, there is no logic for the absolute truth and absolute false value. But in fuzzy logic, there is an intermediate value too present which is partially true and partially false.

For Example: The truth value 0.6 can be assigned to the statement "Fred is happy", because Fred is happy slightly more than most of the time, and the truth value 0.5 can be assigned to the statement "Percy is happy" because Percy is happy half of the time.

Solved Examples

Here are a few solved examples on propositional logic:
Q1: Consider these statements,
P: It will rain today.
Q: I shall go to the party.

Solve these propositions with reference to the above statements:
(i) (~P V~Q)
(ii) (~P ^ Q)
(iii) (P V Q)
Solution:
(i) (~P V~Q) : It will not rain today or I shall not go to the party.
(ii) (~P ^ Q) : It will not rain today and I shall go to the party.
(iii) (P V Q) : It will rain today or I shall go to the party.

Q2: Write the following in symbolic form
"If either Jennie eats pie or Minnie eats cake then Sherry will eat pudding"
Solution:
Let,
p: Jennie eats pie
q: Minnie eats cake
r: Sherry will eat pudding
The symbolic form of the above compound statement is (p v q) -> r.

Q3: Construct a truth table for
(P ->Q) V (~Q ->P)
Solution:

P	**Q**	P ->Q	~Q ->P	P ->Q) V (~Q ->P)
F	F	T	T	T
F	T	T	T	T
T	F	F	T	T
T	T	T	F	T

Basic Logical Laws - Equivalences

Commutative Laws	
$p \lor q \Leftrightarrow q \lor p$	$p \land q \Leftrightarrow q \land p$
Associative Laws	
$(p \lor q) \lor r \Leftrightarrow p \lor (q \lor r)$	$(p \land q) \land r \Leftrightarrow p \land (q \land r)$
Distributive Laws	
$p \land (q \lor r) \Leftrightarrow (p \land q) \lor (p \land r)$	$p \lor (q \land r) \Leftrightarrow (p \lor q) \land (p \lor r)$
Identity Laws	
$p \lor 0 \Leftrightarrow p$	$p \land 1 \Leftrightarrow p$
Negation Laws	
$p \land \neg p \Leftrightarrow 0$	$p \lor \neg p \Leftrightarrow 1$
Idempotent Laws	
$p \lor p \Leftrightarrow$	$p \land p \Leftrightarrow p$
Null Laws	
$p \land 0 \Leftrightarrow 0$	$p \lor 1 \Leftrightarrow 1$
Absorption Laws	
$p \land (p \lor q) \Leftrightarrow p$	$p \lor (p \land q) \Leftrightarrow p$
DeMorgan's Laws	
$\neg(p \lor q) \Leftrightarrow (\neg p) \land (\neg q)$	$\neg(p \land q) \Leftrightarrow (\neg p) \lor (\neg q)$
Involution Laws	
$\neg(\neg p) \Leftrightarrow p$	

Basic Logical Laws - Common Implications and Equivalences

Detachment (AKA Modus Ponens)	$(p \to q) \wedge p \Rightarrow q$
Indirect Reasoning (AKA Modus Tollens)	$(p \to q) \wedge \neg q \Rightarrow \neg p$
Disjunctive Addition	$p \Rightarrow (p \vee q)$
Conjunctive Simplification	$(p \wedge q) \Rightarrow p$ and $(p \wedge q) \Rightarrow q$
Disjunctive Simplification	$(p \vee q) \wedge \neg p \Rightarrow q$ and $(p \vee q) \wedge \neg q \Rightarrow p$
Chain Rule	$(p \to q) \wedge (q \to r) \Rightarrow (p \to r)$
Conditional Equivalence	$p \to q \Leftrightarrow \neg p \vee q$
Biconditional Equivalences	$(p \leftrightarrow q) \Leftrightarrow (p \to q) \wedge (q \to p) \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$
Contrapositive	$(p \to q) \Leftrightarrow (\neg q \to \neg p)$

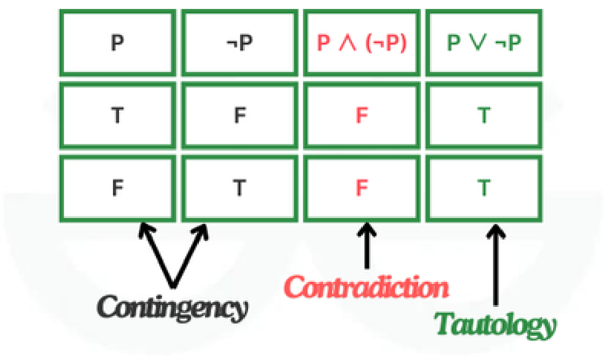
Tautology, Contradiction and Contingency

1.4 Tautology :

Tautology in math is a compound statement that always returns a true value. The individual part of the tautology does not affect the result, which is always true. This concept is fundamental in propositional logic, a branch of logic dealing with propositions that can either be true or false. For example, the statement "It will either snow today or it will not snow today" (p ∨ ¬p) is a tautology because it is always true: either it snows, or it doesn't.

What is Tautology in Math?

The compound statements whose result is always true irrespective of the intermediate result is called the tautology. In other words, the tautology is a compound statement which is always true although the individual statements can be false. For example, the statement "A or not A" (where A is any proposition) is a tautology because it is always true whether A is true or false.



A tautology is a statement in logic and mathematics that is always true regardless of the truth values of its individual components.

Some examples of tautology in maths are:

- Logical Statement: The statement "If it is raining, then it is raining" is a tautology because it is always true.
- Propositional Logic: The expression "P ∨ ¬P" (P or not P) is a classic example of a tautology.
- Set Theory: The statement "A ⊆ A" (A is a subset of A) is always true for any set A, making it a tautology.

Truth Table for Tautology

Consider the logical expression P ∨ ¬ P (P OR NOT P). The truth table for this tautology is as follows:

P	¬P	P v ¬P
T	F	T
F	T	T

Another common example of a tautology would be
(P ∧ Q) ∨ (¬P ∨ ¬Q):

P	Q	¬P	¬Q	P ∧ Q	¬P ∨ ¬Q	(P ∧ Q) ∨ (¬P ∨ ¬Q)
T	T	F	F	T	F	T
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

1.5 Contradiction:

A contradiction is a statement that is always false, regardless of the truth values of the propositions involved. In other words, a contradiction is a statement that cannot be true under any circumstances. For example, the statement P∧¬PP∧¬P is a contradiction because it asserts that P is both true and false at the same time.

Examples of Contradictory Statements

Here are a few examples of contradictory statements:

- P∧¬PP∧¬P
- (P→Q)∧(P∧¬Q)(P→Q)∧(P∧¬Q)
- (P↔Q)∧(P∧¬Q)(P↔Q)∧(P∧¬Q)

These statements are always false, regardless of the truth values of P and Q.

1.6 Contingency:

Contingency is a fundamental concept in propositional logic, referring to a statement that is neither always true nor always false. In other words, a contingent statement is one whose truth value depends on the truth values of its constituent parts.

Definition and Examples of Contingency

A statement is said to be contingent if it is true under some interpretations and false under others. For instance, consider the statement "It is raining or it is not raining." This statement is not contingent because it is always true, regardless of the weather. On the other hand, the statement "It is raining and it is not raining" is no contingent either, as it is always false. However, the statement "It is raining" is contingent because its truth value depends on the actual weather conditions.

To illustrate this further, let's examine the truth table for the statement $P \wedge QP \wedge Q$, where PP represents "It is raining" and QQ represents "The streets are wet."

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

As shown in the truth table, the statement $P \wedge QP \wedge Q$ is contingent because its truth value varies depending on the truth values of P and Q.

Truth Tables and Contingency

Truth tables are a powerful tool for determining the contingency of a statement. By examining the truth table of a statement, we can easily identify whether it is contingent, tautological, or contradictory.

For example, consider the statement (P⋁Q)⋀¬(P⋀Q) (P⋁Q)⋀¬(P⋀Q). To determine whether this statement is contingent, we can construct its truth table:

P	Q	$P \vee Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$(P \vee Q) \wedge \neg(P \wedge Q$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

The truth table reveals that the statement (P⋁Q)⋀¬(P⋀Q) (P⋁Q)⋀¬(P⋀Q) is contingent, as its truth value is not always the same.

Comparison b/w Tautology, Contradiction & Contingency

Property	Tautology	Contradiction	Contingency
Definition	A statement that is always true regardless of the truth values of its components.	A statement that is always false regardless of the truth values of its components.	A statement that can be either true or false depending on the truth values of its components.
Truth Table	All rows in the truth table result in true (T).	All rows in the truth table result in false (F).	Some rows in the truth table result in true (T), and some result in false (F).
Logical Form	"$P \vee \neg P$" (P or not P)	"$P \wedge \neg P$" (P and not P)	"$P \wedge Q$" (P and Q) or "$P \vee Q$" (P or Q) where P and Q have varying truth values.
Example	"It is either raining or it is not raining."	"It is raining and it is not raining."	"It is raining and it is cold." or "It is raining or it is cold."
Usage	Used to validate logical arguments and simplify expressions.	Used to identify impossible situations or logical inconsistencies.	Used to represent realistic scenarios with variable outcomes.

References

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